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Systematic review synthesising the effects of study abroad experience on the development of language fluency

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Abstract

Given the importance of learning contexts in second language (L2) learning, study abroad (SA) has received due attention in the L2 literature, as this context has been proposed to accelerate L2 development. In line with recent efforts to synthesise the effects of SA on L2 learning, the present study reports a meta-analysis of the effects of SA experience on the development of L2 fluency compared to the at-home environment. Our dataset comprised 14 between-group studies, all based on quasi-experimental designs, including a comparison group. The results indicate that the SA context had an overall medium effect (Hedges' g = 0.65, SE = 0.10, 95% CI [0.44, 0.87]) on the development of L2 fluency compared to its at-home counterpart. We also identified several variables related to learner characteristics and SA programme features and further examined their moderating effects on SA. Based on our findings, an in-depth discussion of how these variables are related to the effectiveness of SA was provided, and recommendations for L2 learners and SA researchers involved in SA programmes were proposed.

Keywords: Fluency; learning contexts; meta-analysis; second language; study abroad

Introduction

Considering that learning context is a critical component of second language acquisition (SLA) (Ortega, 2009), one strand of SLA research has examined the extent to which studying abroad (SA) is beneficial for the development of second language (L2) skills and knowledge. An underlying assumption is that 'immersion of the target language in a natural environment, combined with instruction in a classroom setting, can significantly expedite the learning

process and lead to considerable proficiency changes' (Xu, 2019, p. 199). A large volume of L2 research has tested such an assumption, and the field has recently seen several review articles (e.g. Borràs & Llanes, 2021; DeKeyser, 2014; Llanes, 2011; Moore et al., 2021; Sanz, 2014; Tullock & Ortega, 2017) and meta-analyses (e.g. Tseng et al., 2024; Xu, 2019; Yang, 2016) that attest to the immense interest in this topic.

Among the various linguistic dimensions, SA researchers have paid attention to the development of learners' fluency, the target dimension of the current study, which is a component of the widely adopted framework proposed by Wolfe-Quintero et al. (1998) (i.e. complexity, accuracy, and fluency; CAF). Despite different operationalisations of this dimension, our study adopts Lennon's (2000) definition of fluency as 'the rapid, smooth, accurate, lucid, and efficient translation of thought or communicative intention into language under the temporal constraints of on-line processing' (p. 26). The interest in examining this linguistic aspect in SA research is not unexpected, given that oral (or speaking) fluency has been considered synonymous with L2 proficiency by scholars (e.g. Chambers, 1997; Freed, 1995) and non-specialists in L2 learning (Mora et al., 2023).

This study reports a meta-analysis of quasi-experimental SA studies that investigated the effects of SA on L2 fluency development. Our study extends previous synthetic efforts on the effects of SA on language learning (Moore et al., 2021; Tullock & Ortega, 2017; Xu, 2019), which have focused on this issue with different analytical lenses. More specifically, both Tullock and Ortega's (2017) and Xu's (2019) scoping reviews considered within-group (i.e. pre-post design) studies on language learners' interlanguage development via SA experiences, but focused on oral fluency and complexity, respectively, using the CAF framework (Marijuan & Vallejos, 2023; Mora et al., 2023). More recently, Moore et al.'s (2021) systematic review considered between-group studies consisting of both SA and comparison groups to explore whether 'attributing change in language proficiency to that of the SA itself' could be supported based on existing research evidence (p. 4), focusing on those based on undergraduate language learners who experienced more than five weeks of SA. The current study aims to complement and extend these previous synthetic efforts by considering between-group studies of language learners' interlanguage development in fluency (both oral and written) via SA experiences. It also aims to provide evidence-based implications for prospective sojourners and those involved in the administration of SA programmes by considering several variables thought to moderate the effects of SA on L2 learning.

Background

Experimental design

Research on the effects of SA on the development of L2 fluency has adopted either a within-subject design (e.g. Mora & Valls-Ferrer, 2012; Perez-Vidal & Juan-Garau, 2011) or a between-subject design (e.g. Köylü & Tracy-Ventura, 2022; Llanes, 2012). Studies based on the former, commonly known as a one-group pre-test – post-test design in the social sciences (Cook & Campbell, 1979), involve an SA group only and measure participants' fluency before and after the sojourn in the target environment. In contrast, others have additionally included the 'at home' (AH) group as a comparison group and compared the relative effectiveness of SA to learning the target language through formal instruction in the home country.

Some previous reviews on the SA effects (e.g. Tullock & Ortega, 2017) have focused on within-subject studies because of their ability to remove the influence of confounding variables related to individual learner differences. Although the strength of within-subject designs is not underestimated, the present study will focus on between-subject designs, in line with Moore et al. (2021), Tseng et al. (2024), and Yang (2016), given the proposition that the within-subject design 'cannot rule out other possible sources of change, such as the maturation effect, the practice effect, or the Hawthorne effect' (Dörnyei, 2007, p. 117) (see Cook & Campbell, 1979 for a similar explanation). Therefore, the target studies for the present review included both SA and AH (or equivalent) groups and measured their development via the post-test, including fluency-related measures.

Potential moderators

Before conducting a thorough investigation of the present meta-analysis, we reviewed the potential moderators that could influence L2 fluency development. Based on the aforementioned literature, four moderators were identified: (1) programme content, (2) type of residence, (3) learners' pre-sojourn L2 proficiency, and (4) length of SA. In addition to these moderators, we conducted a subgroup analysis of oral versus written fluency (see below for details).

Programme content

SA programmes differ in terms of the content that students take during their sojourn. For instance, some programmes may be geared predominantly towards formal language instruction (e.g. D'Amico, 2010; Serrano et al., 2014), whereas others may include both

language and content courses (e.g. Freed et al., 2004; O'Brien et al., 2007). Indeed, this moderator has been subject to investigation in previous meta-analyses (e.g. Tseng et al., 2024; Xu, 2019), and the present meta-analysis also examined this variable.

Type of residence

Type of residence pertains to the living conditions of students within the target language setting. Living with a host family can provide learners with increased opportunities to practice the target language in authentic conversations with native speakers, which may lead to greater oral proficiency gains (Mancheño, 2008; Vande Berg et al., 2009). However, communicating in an unfamiliar language with host family members can also help learners avoid having conversations with them (e.g. Rivers, 1998; Wilkinson, 1998). Meanwhile, residing in school-based dormitory settings could enable students to engage in both their native language and target language with their peers (e.g. Sasaki, 2007; Serrano et al., 2016). Given the divergent living environments experienced during SA programmes, it can be assumed that the impact of such experiences may influence students' linguistic development; thus, the type of residence was included as a potential moderator in the current meta-analysis.

Learners' pre-sojourn L2 proficiency

An examination of the participant characteristics in previous primary SA studies revealed that students' pre-sojourn language proficiency varied. Language proficiency is an important mediating factor that affects linguistic gains in SA research (DeKeyser, 2014). It has been suggested that students with relatively lower levels of language proficiency may benefit more from the SA experience (e.g. Freed, 1995; Li, 2014) (i.e. have more room for improvement), while it can also be assumed that a certain level of target language proficiency may be a prerequisite for engaging in some level of verbal interaction with other speakers in the SA context. Thus, including this moderator in the meta-analysis is expected to provide evidence for these different propositions.

Length of study abroad

The duration of the SA experience was included as another potential moderator to account for the degree of fluency development. SA research has not been unanimous on the idea that 'the longer SA, the more development'. For example, some scholars (e.g. Juan-Garau, 2015) have argued that a longer sojourn can be more beneficial for learners than a shorter sojourn, whereas others (e.g. Grey et al., 2015; Stevens, 2011) concluded that a shorter SA experience could lead to better linguistic gains among L2 learners. Given that previous meta-analyses,

including this moderator (e.g. Tseng et al., 2024; Varela, 2017; Yang, 2016), have presented conflicting findings, the present study examines whether this variable moderates the extent to which SA experience influences the development of L2 fluency.

Oral and written fluency

It has been proposed that written gains in the SA context may not be as significant as oral gains (e.g. Freed, 1995), as 'sojourns often prioritise spoken language development over written' (Köylü & Tracy-Ventura, 2022, p. 1351). Fluency has been a major linguistic domain in this line of argument, with oral fluency being the most sensitive to language learning experiences via SA (Llanes, 2012), given the immense opportunities for students to engage in oral communication in the target language community. Regarding complexity, one of the components of the CAF framework (Wolfe-Quintero et al., 1998) along with fluency, Xu's (2019) meta-analysis of SA revealed no significant difference in gains between oral and written complexity. In line with Xu's study, this meta-analysis examined the relative effectiveness of SA for oral and written fluency in examining the proposition.

Methods

Research questions

The present meta-analysis is guided by the following three research questions:

RQ1: What are the overall effects of SA on L2 fluency development?

RQ2: To what extent do the effects of SA vary across the identified moderators?

RQ3: To what extent do the effects of SA vary between oral and written fluency?

Literature search and inclusion criteria

We have conducted a literature search and established inclusion criteria as follows to identify potential primary studies: studies had to (1) be written in English; (2) be quantitative studies related to the topic (i.e. SA) and contain dependent variables involving measurements of students' L2 fluency.; (3) have (quasi-) experimental-control design (i.e. including both SA and AH groups) to observe the comparable learning effect after the sojourn between the two learning contexts; and (4) report descriptive statistics to calculate Hedges' *g* for effect sizes.

First, we conducted a keyword search using the following three databases: Web of Science (all databases, including Web of Science Core Collection), Scopus, and ProQuest Dissertation & Theses Global. The combination of keywords used in our research was ['Study abroad'] AND ['fluency' OR 'oral fluency' OR 'written fluency'] AND ['impact' OR 'effect' OR

‘effectiveness’] and this search was limited to abstracts in the English language. We chose this approach to identify primary studies related to our present topic based on the aforementioned four inclusion criteria.

Second, we checked the reference lists from previous studies, including review articles and meta-analyses (e.g. Moore et al., 2021; Tseng et al., 2024; Tullock & Ortega, 2017; Xu, 2019; Yang, 2016) and searched scholarly journals (i.e. *ELT Journal*, *Language Learning*, *Porta Linguarum*, *The Language Learning Journal*, *The Modern Language Journal*, *Studies in Second Language Acquisition*, and *System*) related to our topic.

Third, full articles of the identified studies were carefully reviewed to finally confirm they met our inclusion criteria.

Dataset construction and quality assessment

After examining the collected studies, the first and second authors jointly constructed the dataset. First, we checked whether each of the included studies had more than one independent sample. For example, a study might include three groups based on participants’ age groups: children, adolescents, and adults. If these groups were treated independently in the study, then we could code three samples from this single study.

Second, data on features of study design, context, treatment, outcome, and participants were extracted. After necessary data were extracted from the studies, the quality of the studies were assessed by accounting for their design, scale, missing data, and measurement quality: four elements for quality assessment suggested by Gorard (2024).

Third, we calculated effect sizes. In the current meta-analysis, two group contrasts (i.e. SA and AH) were used to estimate the magnitude of the effects of SA compared to those of AH on L2 fluency development. For this purpose, relevant descriptive statistics, such as sample size, means, and standard deviations related to the performance of the two groups on the post-tests administered after the SA group’s sojourn, were extracted to calculate the effect sizes. For effect size statistics, Hedges’ unbiased estimate of the standardised mean difference effect size (i.e. Hedges’ *g*) was applied to account for differences in sample sizes across studies. It should be noted that studies on this topic (e.g. D’Amico, 2010; Köylü & Tracy-Ventura, 2022) generally included more than one outcome measure. Given the small number of studies identified and their sample sizes, we decided to follow the principle of ‘one effect size per sample’ by averaging the effect sizes on similar outcome measures from one sample. In doing so, we removed any measures for which the homogeneity of the two groups was not confirmed in the pre-test. These effect sizes were then weighted to account for each sample size when

calculating the overall effect size of RQ1. These effect sizes were then used to conduct the meta-regression, through which we aimed to examine the magnitude of the moderating effects of the identified variables per RQ2.

All analyses, including the calculation of the overall effect size and meta-regression, were performed using SPSS 28.0, except for the diagnosis of outliers and influential cases, which was performed using the MAJOR package in Jamovi 2.3.28.

Outlier diagnostics and publication bias

After calculating the effect sizes, we diagnosed potential outliers, given the reporting of previous meta-analyses (e.g. Yang, 2016) on studies with extremely large effect sizes. In light of the suggestions made by Assink and Wibbelink (2016), we initially converted the calculated effect sizes into standardised *z* values and checked if any of these were larger than 3. We then used Jamovi's MAJOR module to examine whether these effect sizes could be identified as an outlier based on mathematical standards, such as studentised deleted residuals, DFFITS values, Cook's distances, and COVRATIO (see Viechtbauer & Cheung, 2010, for details on these values). Additionally, we created a funnel plot to visualise the distribution of effect sizes to check for any signs of publications bias. Then, we employed Egger's regression test based on Sterne et al.'s (2011) recommendation to statistically check whether the visual signs of the funnel plot actually were a significant indication of publication bias among the effect sizes.

Results

Identified studies for meta-anaysis

In total, 14 studies were included in the meta-analysis. The selection process was jointly conducted by the first and third authors. Figure 1 shows the stages of our literature search using the PRISMA flow diagram.

Based on the database search, a total of 64 studies (27 from Web of Science, 24 from Scopus, and 13 from ProQuest Dissertation & Theses Global) were identified. After removing 24 duplicate studies, 40 studies remained. Further, we identified 15 additional studies from the manual search. After removing 30 duplicate studies, a total of 55 studies remained for a further review.

Then, we carefully reviewed these studies to confirm they met our inclusion criteria. A total of 41 studies were excluded for the following reasons: four studies did not cover the SA topic; 33 studies had study designs which did not meet our criteria (e.g. six studies were not empirical, such as review articles and conceptual papers; 27 studies were either qualitative or one-group

design/longitudinal); three studies were not about the target linguistic aspect (i.e. L2 oral/written fluency); and one dissertation study was later published as a journal article, which remained in the current meta-analysis. Finally, 14 studies remained for the meta-analysis.

After reviewing these studies, we found that four studies included multiple independent samples, therefore we treated them as separate substudies. For example, the participants in Llanes and Muñoz (2013) were divided into two groups (child vs. adult), which were considered individual samples. Similarly, in Llanes and Serrano (2017), the participants were divided into three groups (child, adolescent, and adult). In addition, Serrano et al. (2011) divided the AH groups into two groups based on the intensity of the language course provided (AH intensive and AH semi-intensive). Finally, in Köylü and Tracy-Ventura (2022), SA groups were divided into two different contexts (SA vs. ELF [English as a Lingua Franca] SA). Consequently, we identified five more samples, which means that there were 19 samples (14 + 5) among the 14 primary studies. In summary, after considering the data construction of each study, we confirmed 19 (14 + 5) samples (N = 910) from 14 primary studies[1] (see Table 1).

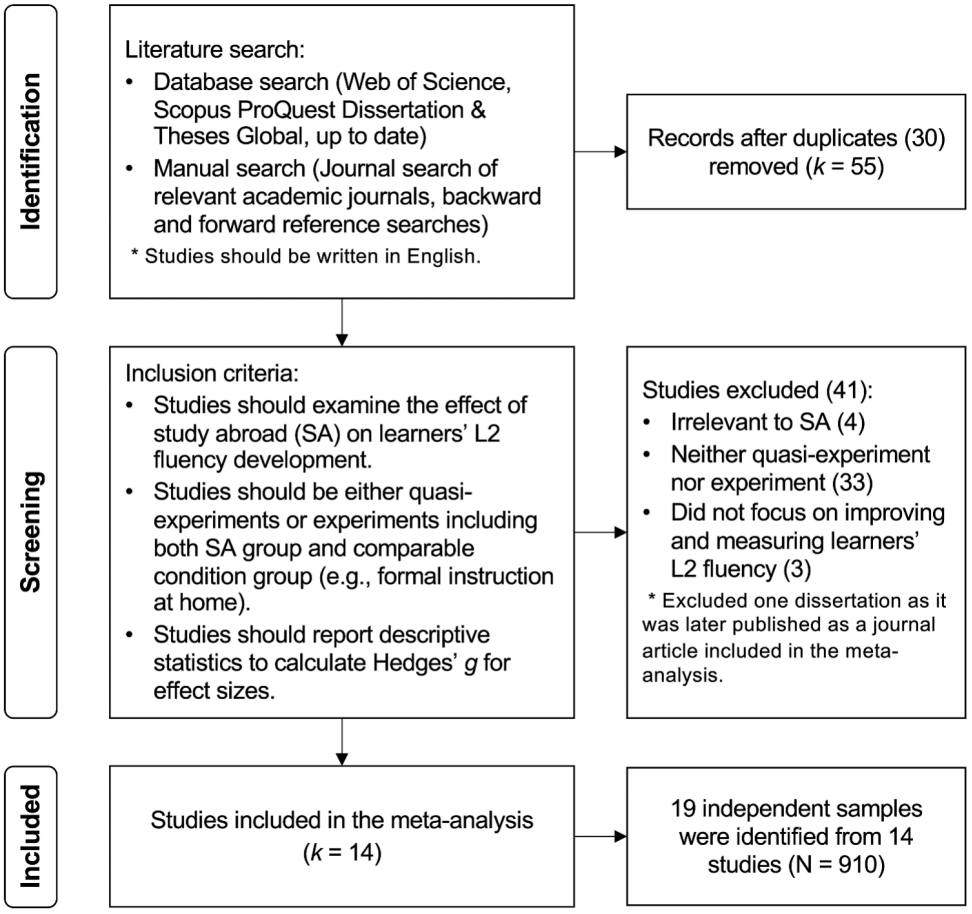


Figure 1. PRISMA flow diagram (Page et al., 2021).

Dataset construction and quality assessment

After extracting data from the collected studies, we coded moderator variables as follows: For the programme content, coding was dependent on the content of the SA programmes received by the participants. The samples were coded as 'formal language' (4 of 19; 21%) when they took only language courses (e.g. grammar, vocabulary and writing) during SA. We coded them as 'Other' (15 of 19; 79%) when participants took content-based courses (e.g. culture, sociology, science) in the target language or took both (i.e. formal language course and content-based course). Regarding the types of residence, we categorised the samples as follows: (1) 'host family' (6 of 19; 32%); (2) 'dormitory' (4 of 19; 21%); and (3) 'mixed' (7 of 19; 37%). Two studies did not provide information related to the residence type (O'Brien et al., 2007; Taguchi, 2013).

Table 1. Sample size of each independent sample in the dataset.

Independent sample	**Sample size**	
	SA group	**AH group**
D'Amico (2010)	9	14
Freed et al. (2004)	8	8
Köylü and Tracy-Ventura (2022)	8	15
Köylü and Tracy-Ventura (2022)	24	15
Llanes (2012)	9	7
Llanes and Muñoz (2013)	39	34
Llanes and Muñoz (2013)	46	20
Llanes and Serrano (2017)	50	44
Llanes and Serrano (2017)	24	15
Llanes and Serrano (2017)	46	18
O'Brien et al. (2007)	25	18
Sasaki (2004)	6	5
Sasaki (2007)	7	6
Segalowitz and Freed (2004)	22	18
Serrano et al. (2011)	24	69
Serrano et al. (2011)	25	37
Serrano et al. (2014)	54	58
Serrano et al. (2016)	16	25
Taguchi (2013)	22	20
Total: 19 independent samples (k = 19)	464	446

Learners' pre-sojourn L2 proficiency was coded, depending on the identified studies' authors' description and also by referring to the participants' number of previous years of L2 learning; the sample was categorised as (1) 'beginner/pre-intermediate' (8 of 19; 42%), (2) 'intermediate and above' (10 of 19; 52%), and (3) 'mixed' (1 of 19; 6%). The length of the SA was coded as a categorical variable (i.e. >15 weeks vs. <15 weeks) in our meta-analysis. The majority of the

studies directly reported the length using units such as months or weeks. In the present study, we followed previous meta-analyses (e.g. Tseng et al., 2024; Xu, 2019; Yang, 2016), which coded length as a unit of weeks. Table 2 summarises the key characteristics of the collected samples.

After the data extraction, the trustworthiness of each sample was assessed to ensure their rigour as (quasi-) experiments, with respect to their design, scale, missing data, and measurement quality (see Gorard, 2024, for rationales for choosing these four elements for this purpose). Following Gorard's (2024) guidelines, the design was assessed to determine if the designs were strong, adequate, weak, very weak, or no consideration for the research questions. For the scale, we assessed whether there were large, adequate, small, very small, or trivial numbers of cases. For the missing data, we assessed whether the samples had a minimal quantity, some, a moderate quantity, high levels, or a huge quantity of missing data. Lastly, for the measurement quality, we assessed whether the samples had standardised, independent, and accurate measurements; standardised and independent measurements with some errors; non-standardised/independent measurements with errors; weak measures with a high level of error; or very weak measures. For each value, we gave ratings from four to zero. When a sample had no consideration of design, a trivial scale, a huge amount of missing data, or very weak measures, we gave a rating of zero. We decided that if a sample had an average score of three or higher, it was considered appropriate for inclusion in the meta-analysis. As a result, we found that all samples successfully passed our quality assessment. Table 3 summarises the results of the quality assessment.

Table 2. Key characteristics of the collected samples.

Independent Sample	Programme Content	Type of Residence	L2 Proficiency	Length of SA
D'Amico (2010)	Formal language	Host family	Intermediate and above	<15 weeks
Freed et al. (2004)	Other	Mixed	Intermediate and above	<15 weeks
Köylü and Tracy-Ventura (2022)	Other	Mixed	Intermediate and above	>15 weeks
Köylü and Tracy-Ventura (2022)	Other	Mixed	Beginner/Pre-intermediate	>15 weeks
Llanes (2012)	Other	Host family	Beginner/Pre-intermediate	<15 weeks
Llanes and Muñoz (2013)	Other	Host family	Beginner/Pre-intermediate	<15 weeks
Llanes and Muñoz (2013)	Other	Mixed	Intermediate and above	<15 weeks
Llanes and Serrano (2017)	Other	Host family	Beginner/Pre-intermediate	<15 weeks
Llanes and Serrano (2017)	Formal language	Host family	Intermediate and above	<15 weeks
Llanes and Serrano (2017)	Formal language	Mixed	Intermediate and above	<15 weeks
O'Brien et al. (2007)	Other	Not specified	Mixed	<15 weeks

Independent Sample	Programme Content	Type of Residence	L2 Proficiency	Length of SA
Sasaki (2004)	Other	Dormitory	Beginner/Pre-intermediate	>15 weeks
Sasaki (2007)	Other	Dormitory	Beginner/Pre-intermediate	>15 weeks
Segalowitz and Freed (2004)	Other	Host family	Beginner/Pre-intermediate	<15 weeks
Serrano et al. (2011)	Other	Mixed	Intermediate and above	<15 weeks
Serrano et al. (2011)	Other	Mixed	Beginner/Pre-intermediate	<15 weeks
Serrano et al. (2014)	Formal language	Dormitory	Intermediate and above	<15 weeks
Serrano et al. (2016)	Other	Dormitory	Intermediate and above	<15 weeks
Taguchi (2013)	Other	Not specified	Intermediate and above	>15 weeks

Outlier diagnostics and publication bias

After constructing the dataset, including extracting the data from the collected studies for coding moderator variables and assessing the methodological qualities, we calculated effect sizes for each sample. Thus, in total 19 effect sizes were finally computed. Before delving into further data analyses to answer the research questions, we diagnosed any outliers and any signs of publication bias. First, the calculated effect sizes were converted into standardised *z* values, revealing that O'Brien et al.'s (2007) study could be a potential outlier, with a *z* value of 3.08. Second, using Jamovi's MAJOR module in the first round of analysis, the effect size of O'Brien et al.'s (2007) study was identified as an outlier. After removing O'Brien et al., we ran the procedure again, and Segalowitz and Freed (2004) were further identified as outliers. Thus, these two studies were removed from further analysis. No other outliers were detected after removing these two samples, as shown in Figure 2. The remaining effect sizes' values for studentised deleted residuals, DFFITS values, Cook's distances, and COVRATIO (see Viechtbauer & Cheung, 2010, for details on these values) were within the mathematically appropriate ranges, as indicated by the four graphs in Figure 2.

As the next step, we illustrated a funnel plot and ran Egger's regression test. As shown in Figure 3, the majority of effect sizes fell between the funnel lines and were distributed symmetrically around the vertical line, except for three effect sizes. Given the subjective nature of judging publication bias based on the funnel plot and the number of samples included in the dataset (between 10 and 25), we further employed Egger's regression test based on Sterne et al.'s (2011) recommendation. Egger's test was not statistically significant, showing no significant indication of publication bias among the effect sizes (p = .59).

Table 3. Methodological quality of the included samples.

Independent Sample	Design	Scale	Missing Data	Measurement Quality
D'Amico (2010)	Adequate	Small	Some	Standardised, independent, and accurate measurements
Freed et al. (2004)	Weak	Small	Minimal	Standardised, independent, and accurate measurements
Köylü and Tracy-Ventura (2022)	Adequate	Small	Minimal	Standardised, independent, and accurate measurements
Köylü and Tracy-Ventura (2022)	Adequate	Small	Minimal	Standardised, independent, and accurate measurements
Llanes (2012)	Adequate	Small	Minimal	Standardised, independent, and accurate measurements
Llanes and Muñoz (2013)	Adequate	Adequate	Minimal	Standardised, independent, and accurate measurements
Llanes and Muñoz (2013)	Adequate	Adequate	Minimal	Standardised, independent, and accurate measurements
Llanes and Serrano (2017)	Adequate	Adequate	Minimal	Standardised, independent, and accurate measurements
Llanes and Serrano (2017)	Adequate	Small	Minimal	Standardised, independent, and accurate measurements
Llanes and Serrano (2017)	Adequate	Adequate	Minimal	Standardised, independent, and accurate measurements
O'Brien et al. (2007)	Weak	Small	Minimal	Standardised, independent, and accurate measurements
Sasaki (2004)	Weak	Small	Minimal	Standardised, independent, and accurate measurements
Sasaki (2007)	Weak	Small	Minimal	Standardised, independent, and accurate measurements
Segalowitz and Freed (2004)	Adequate	Small	Minimal	Standardised, independent, and accurate measurements
Serrano et al. (2011)	Weak	Adequate	Some	Standardised, independent, and accurate measurements
Serrano et al. (2011)	Weak	Adequate	Minimal	Standardised, independent, and accurate measurements
Serrano et al. (2014)	Weak	Adequate	Minimal	Standardised, independent, and accurate measurements
Serrano et al. (2016)	Adequate	Small	Minimal	Standardised, independent, and accurate measurements
Taguchi (2013)	Weak	Small	Minimal	Standardised, independent, and accurate measurements

Overall effect size

In total, we had 17 effect sizes for further data analyses. Figure 4 illustrates the forest plot for these 17 effect sizes, which range from −0.21 to 1.64. Considering their 95% confidence intervals (CIs), the effect sizes can be categorised into three types. First, Serrano et al.'s (2016) effect size of −0.21 included zero in its CI, indicating that this sample did not demonstrate any

significant negative impact of SA on L2 fluency. Second, a total of six samples – Serrano et al. (2011), Freed et al. (2004), Köylü and Tracy-Ventura (2022), Sasaki (2004), Llanes and Serrano (2017), and Llanes (2012) – had positive effect sizes, but their CIs included zero, indicating that these samples did not demonstrate any significant positive impact. Third, all the other significant effect sizes (i.e. those without zero in their 95% CIs) showed that the impact of SA on L2 fluency was positive, ranging from 0.49 to 1.64. When synthesising all effect sizes across the three types to obtain the overall average effect size, it was 0.65 (Hedges' g) (SE = 0.10; z = 6.32; $p < .001$; 95% CI [0.44, 0.87]). In light of Plonsky and Oswald's (2014) benchmarks for effect size in second language research (between-group standardised mean difference), this overall effect size can be considered medium.

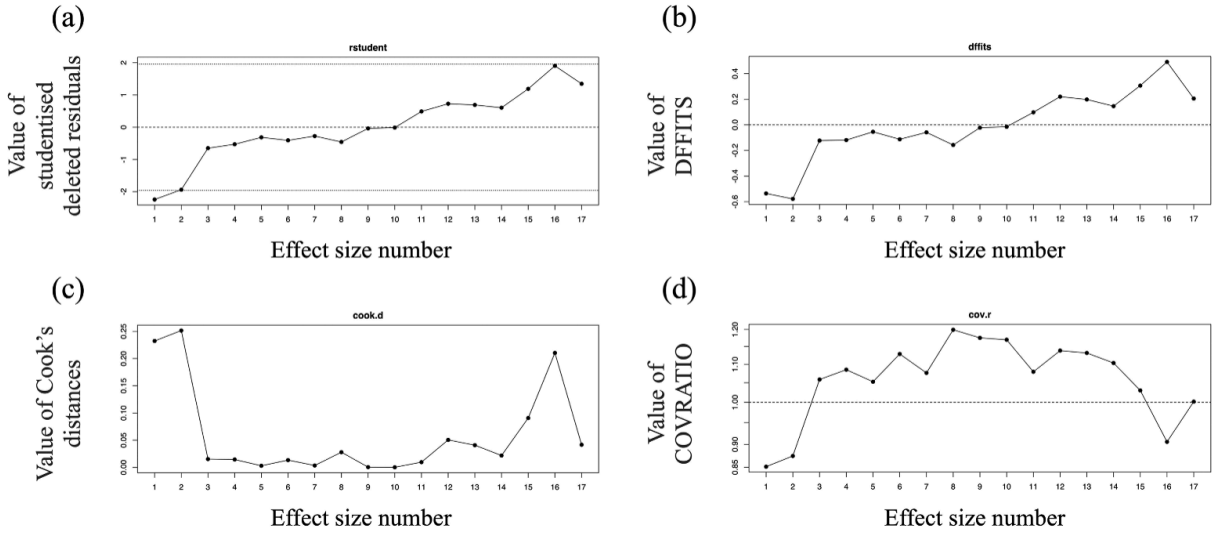


Figure 2. Plot of the (a) studentised deleted residuals, (b) DFFITS values, (c) Cook's distances, and (d) COVRATIO values for 17 effect sizes of post-tests for the comparison of SA versus AH.

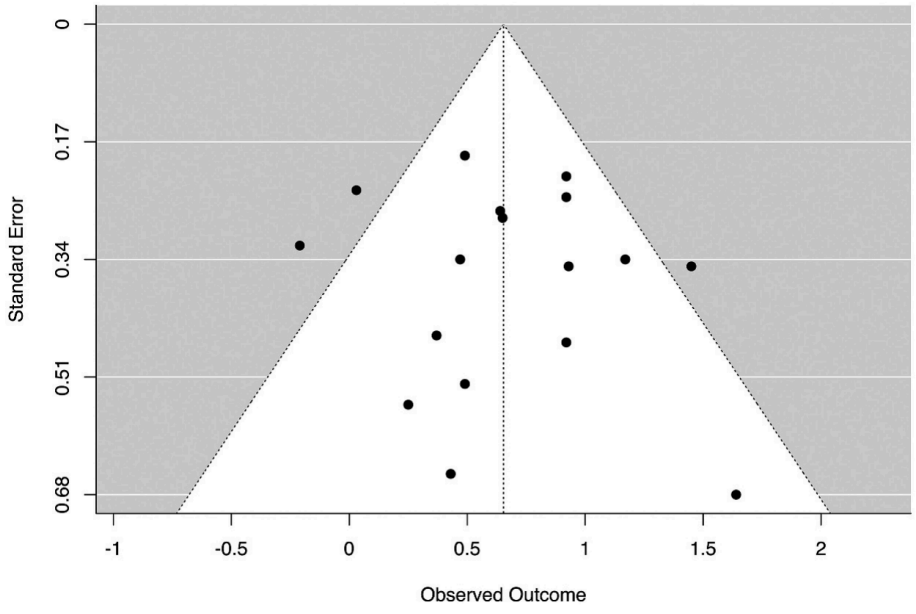


Figure 3. Funnel plot for the post-test results of the comparison SA versus AH on fluency.

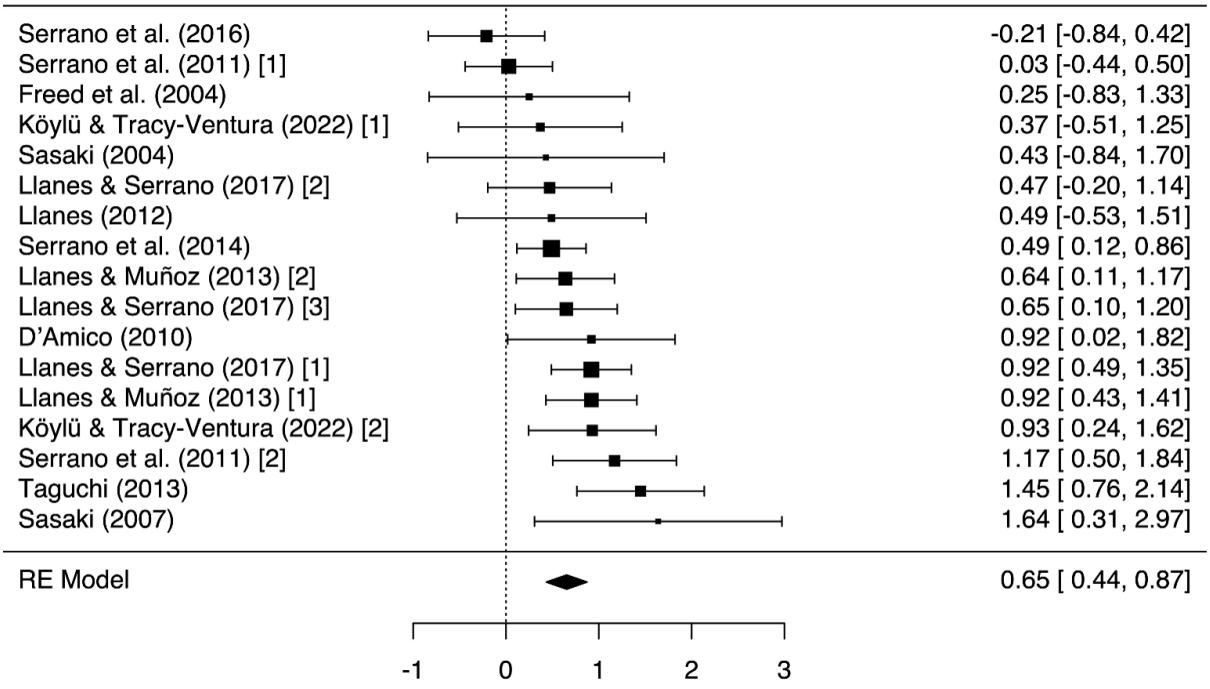


Figure 4. Forest plot for the post-test results of the comparison SA versus AH on fluency.

Table 4. Multiple meta-regression with four moderator variables.

	Moderator variables	**# of effect size**	**Mean effect size**	***SE***	***p***
Programme content	Formal language	4	.63	0.23	.048
	Mixed	13	.69	(reference)	
L2 proficiency	Beginner/Pre-intermediate	7	.93	(reference)	
	Intermediate and above	10	.51	0.26	.012
Residence type	Host Family	5	.74	0.25	.57
	Dormitory	4	.59	0.22	.19
	Mixed	7	.58	(reference)	
SA length	Over 15 weeks	5	.96	(reference)	
	Under 15 weeks	12	.56	.32	.78

Note: Reference level of each moderator variable is marked '(reference)'.

Moderator analysis

To answer the second research question, we conducted a meta-regression analysis. For this analysis, the publication years were included as control variables.

For the meta-regression analysis, four moderators were included (i.e. programme content, learners' pre-sojourn L2 proficiency, residence type, and length of SA; see Table 4). The results revealed that SA had a larger effect on beginner and pre-intermediate learners (*SE* = 0.26; $p < .05$; 95% CI [−1.38, −0.22]) and was more effective for learners when classroom instruction was combined with both formal language courses and other types of courses

(e.g. content) than formal language courses only (*SE* = 0.23; $p < .05$; 95% CI [0.004, 1.04]). However, the type of residence and length of SA were not significant moderators, although staying with host families and longer SA experience resulted in larger effect sizes.

Oral versus written fluency

In addition, we conducted a subgroup analysis to compare the mean effect size for oral and written fluency as an attempt to answer the third research question. It should be noted here that we could not adhere to the 'one effect size per sample' principle for the third research question, which concerned the relative effects of SA on oral and written fluency. This was because some studies (e.g. Llanes & Muñoz, 2013; Serrano et al., 2011) included the measures related to both oral and written fluency. Therefore, for such studies, each effect size related to oral and written fluency was calculated by dividing the sample size by half, which might have otherwise resulted in an underestimation of the standard error. Therefore, the results related to RQ3 should be interpreted with caution and considered exploratory.

According to the results, the average effect size for oral fluency was 0.76 (*SE* = .15; $p < .001$; 95% CI [.48, 1.05]) and that for written fluency was 0.41 (*SE* = .17; $p = .01$; 95% CI [0.08, 0.73]), and the difference between them was marginally significant ($p = .05$). The variance within each fluency measure did not indicate significant heterogeneity (QW = 17.55; $p = .09$ for oral fluency and QW = 7.88; $p = .34$ for written fluency).

Discussion

Overall effectiveness of SA for the development of L2 fluency and quality of the included studies

In line with the previous meta-analyses (e.g. Hirai, 2018; Xu, 2019; Yang, 2016), the present study's finding suggests that SA experience brought about considerable gains in language learning – fluency in our case – compared to its AH counterpart, with the overall average effect size being 0.65 (Hedges' *g*), indicating the medium effect size. The significant development in fluency observed here can be accounted for by the characteristics of SA environments, which provide immense opportunities to engage in communication with the native speaker community, leading to the automatisation of target language skills (DeKeyser, 2007). Given the close relationship between automaticity and fluency in L2 performance (Tavakoli, 2019), SA experience could contribute to the development of target language fluency. Considering previous meta-analyses of SA, the average effect size found in the

present study was similar to those reported by Yang (2016) (d = 0.75), Hirai (2018) (g = 0.56), and Tseng et al. (2024) (g = 0.87). This average effect size was larger than that in Xu's (2019) meta-analysis of the effectiveness of SA in the development of L2 complexity (d = 0.37).

Given the nature of the present study (i.e. systematic review), it is also worth discussing the study quality of the included studies in light of Gorard's (2024) guidelines. First, in terms of measurement quality, all studies were of high quality, including standardised, independent, and accurate measurements (e.g. total number of words per total time and pruned syllables per minute). The studies also had no missing data, with the exception of two studies (D'Amico, 2010; Serrano et al., 2011), thus also demonstrating high quality in this regard. On the other hand, the sample sizes were generally small (see Table 1 for details), with only one of the included samples having more than 100 participants, both SA and AH groups combined. This is one of the most serious problems of SA research, potentially leading to a Type II error (Moore et al., 2021), although the nature of the treatment and its characteristics (e.g. cost, eligibility) may partly account for such small-scale SA research. Finally, none of the studies could be considered strong in terms of study design, as randomisation was not employed in any of the included studies (even at the class level), although some studies (e.g. Köylü & Tracy-Ventura, 2022; Serrano et al., 2016) made efforts to confirm the homogeneity of the SA and AH groups by administering the pre-tests. Again, this aspect (i.e. study design) of SA research is another weakness, but may be difficult to address given that students generally volunteer for their study abroad experience (rather than being directed to do so by teachers or schools).

Role of moderator variables

The results from the further moderator analysis indicated that the overall effectiveness of fluency development during a sojourn was affected by two factors (i.e. programme content and learners' pre-sojourn L2 proficiency). These findings are worth mentioning in-depth to provide evidence-based implications for further SA studies.

Regarding programme content, our results revealed that developments in L2 fluency may be higher when the SA programmes include both formal language courses and other types of courses (e.g. content) than when they are based on formal language courses. This finding concurs with Tseng et al.'s (2024) finding that programmes consisting of both types of courses (i.e. mixed) showed the highest mean effect size. The finding of the current meta-analysis that programmes consisting of language and content courses showed larger effect size may be accounted for in terms of the characteristics of the target linguistic dimension of this study: fluency. That is, given that learners can engage in purposeful communication when taking content courses in the target language (i.e. content and language-integrated learning; Dallinger

et al., 2016; Lyster & Ruiz de Zarobe, 2018), they could likely develop their fluency effectively under such conditions. Based on this finding, those involved in SA programmes may consider integrating content-based courses to enhance learners' fluency.

Another significant moderator was learners' pre-sojourn L2 proficiency level, which confirms the arguments of earlier SA studies that lower-level students may benefit more from SA experiences than their higher-level counterparts (e.g. Brecht & Robinson, 1995; Lapkin et al., 1995; Li, 2014). Given that 'it is more difficult to move up one proficiency sublevel as one progresses up the proficiency scale', as observed in students who study abroad (Baker-Smemoe et al., 2014, p. 477), linguistic gains related to fluency may also show such a pattern. However, our findings should not be interpreted as the SA experience not being beneficial for higher-level students in terms of developing L2 fluency; it simply indicates that more improvements can be expected for relatively lower-level students. Therefore, students and their parents should refer to this finding before making decisions about their sojourns.

Before proceeding to the next section, another finding is the effect size related to oral and written fluency. Our findings revealed that learners could develop more oral than written fluency, as proposed in previous studies (Freed et al., 2003; Llanes, 2012). This finding is not unexpected because SA contexts provide immense opportunities to engage in oral communication in the native speech community (Freed et al., 2004; Serrano et al., 2011). Given this finding, teaching practitioners in SA programmes may incorporate fluency-developing writing activities (i.e. writing on free topics) to balance the development of fluency in the two modes.

Limitations and future research

This section discusses the limitations of this meta-analysis. First, the number of studies included in the dataset and the total number of participants were small despite our efforts to identify as many primary studies as possible through various channels. Unfortunately, this has made the interpretation of some of our findings difficult, and the use of approaches such as multilevel analysis is not feasible. Indeed, because we employed a 'one effect size per sample' approach, the characteristics of our dataset did not allow us to examine the effects of SA on the development of different (sub-)types of fluency, such as utterance fluency, breakdown fluency, and speed fluency. Thus, we highlighted the need for a larger number of primary studies to make such analyses more feasible. Second, due to the small number of effect sizes, some of the moderators were coded as binary in our multiple meta-regression. For example, length of SA, which is actually a scale variable, was necessarily coded as a categorical variable for the purposes of this analysis, with a cut-off point of 15 weeks. Such a decision involved a significant oversimplification of the data and may limit our implications in that

differences in a few weeks of study abroad could, in reality, greatly affect language learning. Lastly, it should be noted that, although some of the moderators have been identified as important in terms of their role in the effectiveness of SA in the previous meta-analyses (e.g. Tseng et al., 2024; Xu, 2019), we could not include them in our analysis due to lack of detailed descriptions. One of these moderators is pre-programme training (e.g. D'Amico, 2010; Freed et al., 2004), which could play a significant role in the effectiveness of SA. Another is the learning experience associated with the AH context, the inclusion of which in future meta-analyses would enable one to gauge the effects of the SA experience more accurately. Overall, we encourage future SA researchers to provide more detailed descriptions of SA and AH programmes in the methodology section.

Conclusion

The present meta-analysis examined the overall effectiveness of SA in the development of L2 fluency, which is arguably one of the most important linguistic dimensions for L2 learners and their parents, in comparison to AH contexts. We also identified and examined a set of potential moderators in this regard. The results show that the SA context was more effective than its AH counterpart in terms of this linguistic dimension, but some variables, such as SA programme content and learners' L2 proficiency, may moderate its effects to different degrees. Furthermore, the SA context may be more effective in improving oral fluency than written fluency.

This study contributes to SA research in two ways. First, as mentioned in the introduction, it adds crucial synthetic evidence to recent SA research that has examined the development of L2 fluency through SA within the CAF framework (e.g. Marijuan & Vallejos, 2023; Mora et al., 2023; Xu, 2019). Second, our study complements and also provides stronger evidence than previous meta-analyses based solely on within-group studies (e.g. Tullock & Ortega, 2017) on the effectiveness of SA for L2 fluency. Nevertheless, we encourage SA researchers to conduct more between-group studies focusing on different subcategories of L2 fluency with a more detailed description of SA contexts, which would allow for a meta-analysis using a more sophisticated statistical approach.

Note

1. We did not consider immersion programme (IM) group in B. Freed et al. (2004) and Group 1 in Taguchi (2013), given the former was exposed to the environment that is very different from SA contexts, and the latter was not comparable to the SA group in terms of its L2 proficiency level.

Disclosure statement

No potential conflict of interest was reported by the authors.

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